AutoDESN remediation

NHS1 nonlinear, seasonal and heteroscedastic series | 24 October 2026

Execution and validation checkpoint

Outcome	Verified status
Implementation	Opt-in augmented readouts, period averaging, normalized grouped RHS, conditional SV, filtering, exact proposals and causal recursion are implemented.
Numerical correctness	478 source assertions pass; installed test results are reported separately. Independent reference outcomes and their scopes are reported separately.
Production inference	No production posterior is yet scientifically qualified. Individually passing R-hat and ESS are insufficient when architecture-frequency or cross-budget gates fail.
Predictive comparison	No new full rolling-origin forecasts or independent confirmation scores are claimed. Historical scores are shown only as unchanged context.

The work is not complete from start to finish. The distinction is deliberate: numerical identities, computational adequacy and predictive improvement are different requirements. Unfinished experiments cannot supply forecast scores or evidence of improvement.

Repository: AntonioAPDL/exdqlm. Feature branch: feature/autodesn-remediation-20261023. Draft pull request: https://github.com/AntonioAPDL/exdqlm/pull/5.

This report is generated from the output files listed in report_inputs.json. It preserves failed runs and source/checkpoint identities; scripts alone are never counted as executed experiments.

Frozen statistical and forecasting contracts

Component	Executed reference convention
Data protocol	500 washout observations; 2,000 scored training observations; 500 held out. Fit once at 2500; origins 2500:2970; horizons 1:30.
Feature geometry	Original 456 reservoir definitions, 426 admitted. Training-only column centers and sample SDs; no square-root dimension normalization. Conditioning-prefix response/input transform is frozen.
Intercept and noise	Independent intercept Normal(0, variance 10); observation variance IG(shape 3, rate 1).
Normalized RHS	tau = 0.001 * u; u and each local scale are unit half-Cauchy. Slab variance IG(2,2): slab_scale=1, slab_df=4. Coefficient variance = 1 / (1/(tau*local)^2 + 1/slab_variance).
Mean families	F1 adds lags 1:90. F2 adds period-averaged Fourier terms over integers 20:120, up to 16 non-Nyquist harmonics. Flat and separately named grouped priors are supported.
Grouped prior	Independent lag/reservoir global scales; one shared slab; Gaussian calendar variance 0.5^2/[1+(harmonic/4)^4].
Conditional SV	Standardized stationary AR(1) state; (phi+1)/2 ~ Beta(10,1.5); omega ~ HalfNormal(0.5). Optional four-harmonic centered log variance.

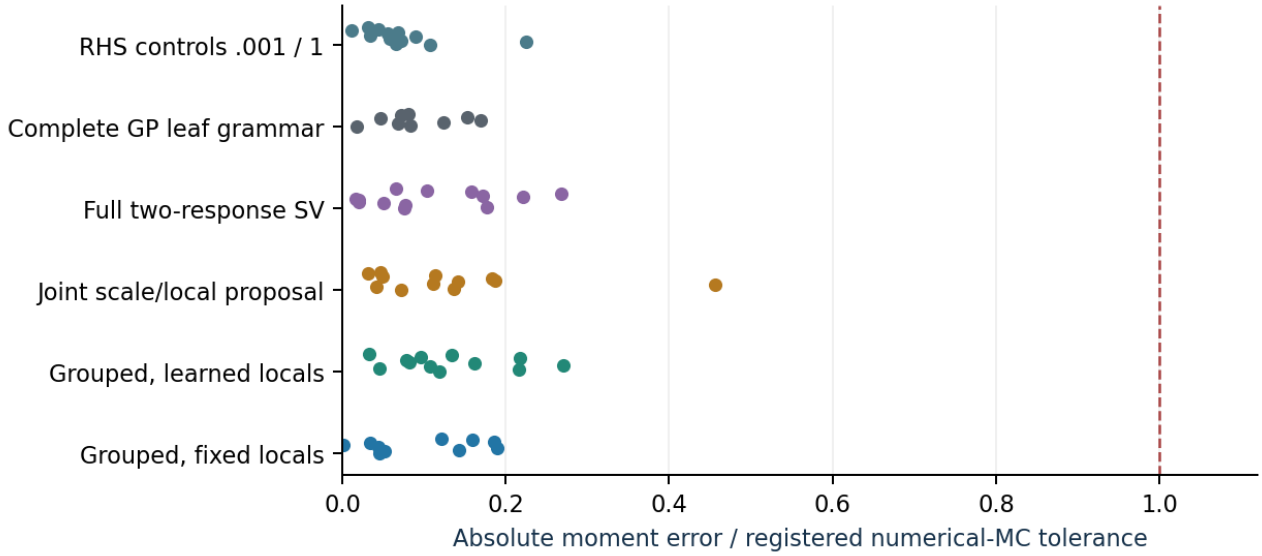
Definition audit: all336 deep candidates use identity bridges; no PCA is fitted. The builder can draw seeded random reducers for other dimensions. The plan correction changes no model or prior. Prefix folds retain those maps and refit training scalers; see PLAN_CORRECTIONS_20261024.md.

Every issued path holds its joint architecture, period and static parameters fixed. Simulated responses update future lags and reservoir states. Actual observations update states only between origins. Conditional SV filtering retains frozen outer parameter weights and replays training uncertainty; it is not full sequential parameter reweighting.

The normalized RHS is not the augmented rhs_ns target. Coefficient-independent noise scaling, all Gaussian determinants, inactive proper priors and exact Hastings corrections are retained. See MATH.md for the joint densities and partial-collapse order.

Independent posterior validation

Conditional reference	Quantities	Max R-hat	Min bulk / tail ESS	Check
Grouped, fixed locals	10	1.000291	22,775 / 28,694	PASS
Grouped, learned locals	12	1.002188	1,830 / 1,988	PASS
Joint scale/local proposal	12	1.002507	1,959 / 1,757	PASS
Full two-response SV	13	1.000482	7,756 / 6,979	PASS
Complete GP leaf grammar	9	1.000505	9,587 / 4,487	PASS
RHS controls .001 / 1	12	1.000693	2,018 / 1,872	PASS



References use dense response-space Gaussian covariances and integration over normalized scale priors. The learned-local check integrates seven dimensions and tests nonuniform architecture/period proposals, variable active dimensions and the added reservoir-local proposal. SV uses independent full two-response quadrature, not only scalar conditional identities.

The initial grouped RHS references use tau0=0.2. The additional seven-dimensional check uses the requested tau0=0.001, slab_scale=1, slab_df=4, with a proper intercept and unknown noise. The two-response SV target is intercept-only. These conditional checks do not certify the complete production workflow.

The GP posterior-integration reference covers all three supported leaves and all four latent parameters under the pinned default depth-one grammar. It checks normalized scores and prior-independence MH; it does not qualify depth-four SMC.

An additional finite conditional check enumerates all 30 default depth-two syntactic trees, including nine change-point trees and ordered sum/product pairs, across five latent fixtures. Independent covariances, normalized active/inactive prior constants and conditional architecture probabilities agree. This checks multiplicities at that depth; it does not integrate all continuous parameters or prove deep-tree exploration.

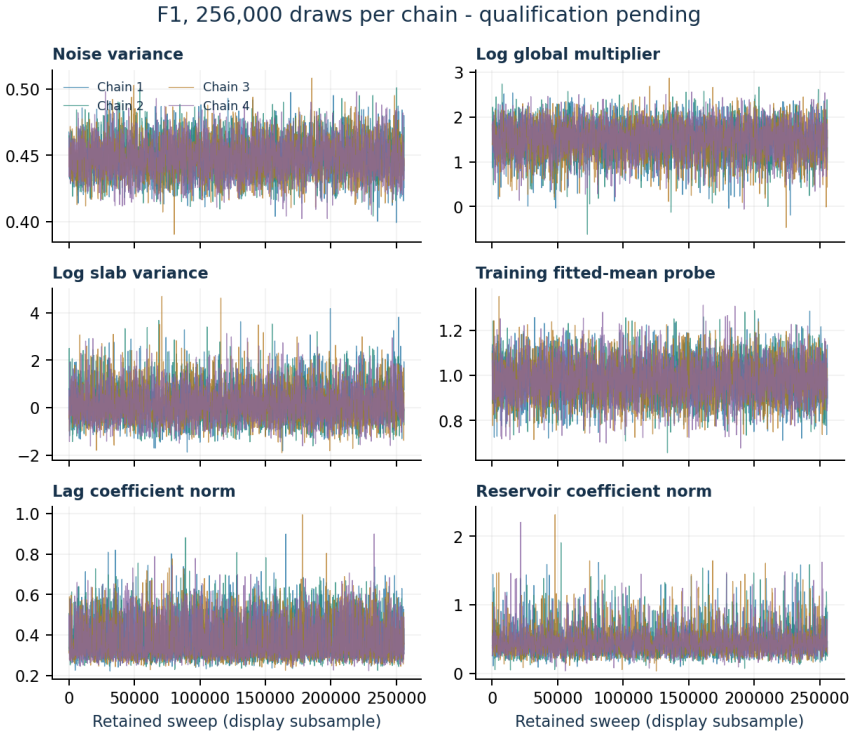
The AutoDESN extension source suite has 29 tests and 478 passing assertions; its installed suite records 478 passing assertions. Earlier legacy regression evidence records 804 passes with one separately covered context skip. Passing these checks is not proof of convergence on the 426-reservoir bank.

The broader GitHub package check on commit bf347f3 failed: 155 test failures, 50 test warnings and four package-check errors. The AutoDESN extension contexts have no failure headers in that log. Packaging of repository-only paths, missing example data, repository integration fixtures and other legacy regression failures remain under investigation; the scoped extension results do not establish package-wide success.

Two stale fixture contexts have since been repaired and all 47 targeted assertions pass. The transfer-function example now executes using shipped climate data; repository-only config is excluded from the package tarball. A complete fresh package check remains separate.

Production convergence: all gates still apply

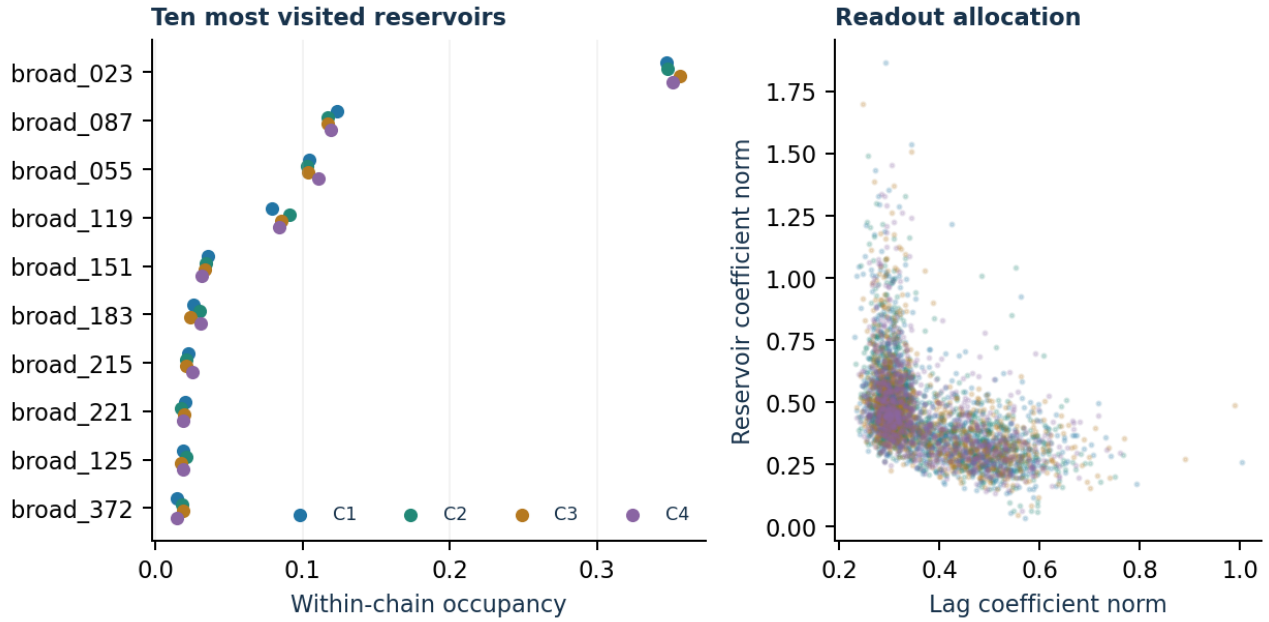
F1 run	R-hat max	ESS min	Largest chain frequency range	Screen
Baseline 32,000	1.0138	305	0.0580	FAIL
Joint full-vector 32,000	1.0068	1,033	0.0549	FAIL
Local move 32,000	1.0122	441	0.0464	FAIL
Baseline 128,000	1.0024	990	0.0528	FAIL
Joint full-vector 128,000	1.0014	3,693	0.0407	FAIL
Joint full-vector 256,000	1.0009	4,326	0.0204	pass



Registered gates: rank/folded R-hat <=1.01, bulk/tail ESS >=400, major-category probability MCSE <=0.01, chain frequency ranges <=0.03. Two successive qualified fresh budgets must also agree within practical and Monte Carlo tolerances. No gate is waived because a different diagnostic passes.

The 256k joint run is the first individual level to pass. All 73 training-probe, predictive-quantile and major-probability comparisons against 128k pass their practical/MC tolerances, but the cross-budget certificate remains false because 128k fails its own frequency screen. A fresh 512k level with 128k warmup is registered before execution; it keeps the same frozen proposals, prior and target.

Architecture exploration and shrinkage



These occupancies and coefficient norms use the latest complete run shown above. Full posterior qualification remains pending even when an individual screen passes. Distinct reservoir modes can have similar fitted functions; a small continuous-parameter R-hat alone does not establish adequate architecture probabilities.

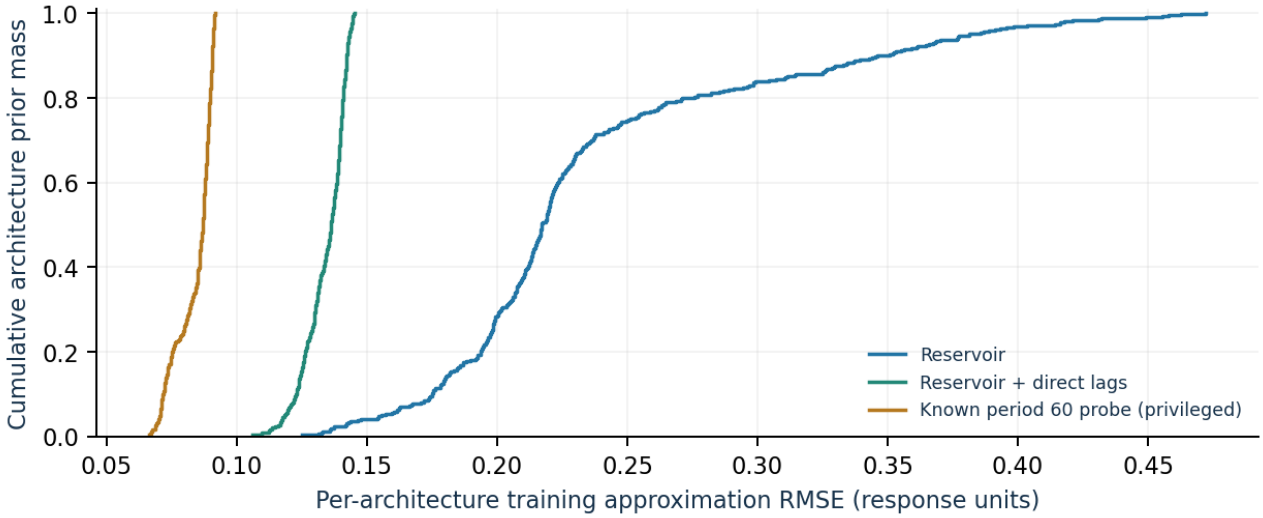
The RHS prior is continuous; exact zeros are not its sparsity mechanism. Coefficient norms and threshold counts depend on the standardized, correlated basis and do not measure a selected-variable count. Conditional group EDF is recorded at the Gaussian substep; a separate audit recomputes it at matching saved scales.

The matching-state audit finds a median 79.6% of standardized-feature slopes below 0.01 response SD, and median 5 above 0.1 response SD. Median actual global scale is 0.00450, since the multiplier is sampled despite tau0=.001. Median slab SD is 1.040; slab_scale=1 is a hyperprior scale. This documents strong shrinkage with retained effects, not adequate predictive fit.

The optional computational repair proposes active reservoir-local scales from a normalized prior/kernel-product mixture and inactive slots from their proper priors. It restores marginalized auxiliaries, then redraws coefficients. Every reservoir retains support. The proposal is frozen from training-only checkpoints; it changes computation, not tau0, the slab, the likelihood or architecture priors.

The 128k-trained proposal has kernel estimates for 27 reservoirs and uses exact prior proposals for 399 others. Scarce training samples do not remove a reservoir.

Representation: a training-only capacity probe



Feature space	Median rank / columns	Prior-weighted mean RMSE	Best RMSE
Reservoir only	61 / 61	0.236	0.125
Add direct lags 1:90	151 / 151	0.134	0.106
Add known calendar60 (privileged)	183 / 183	0.084	0.066

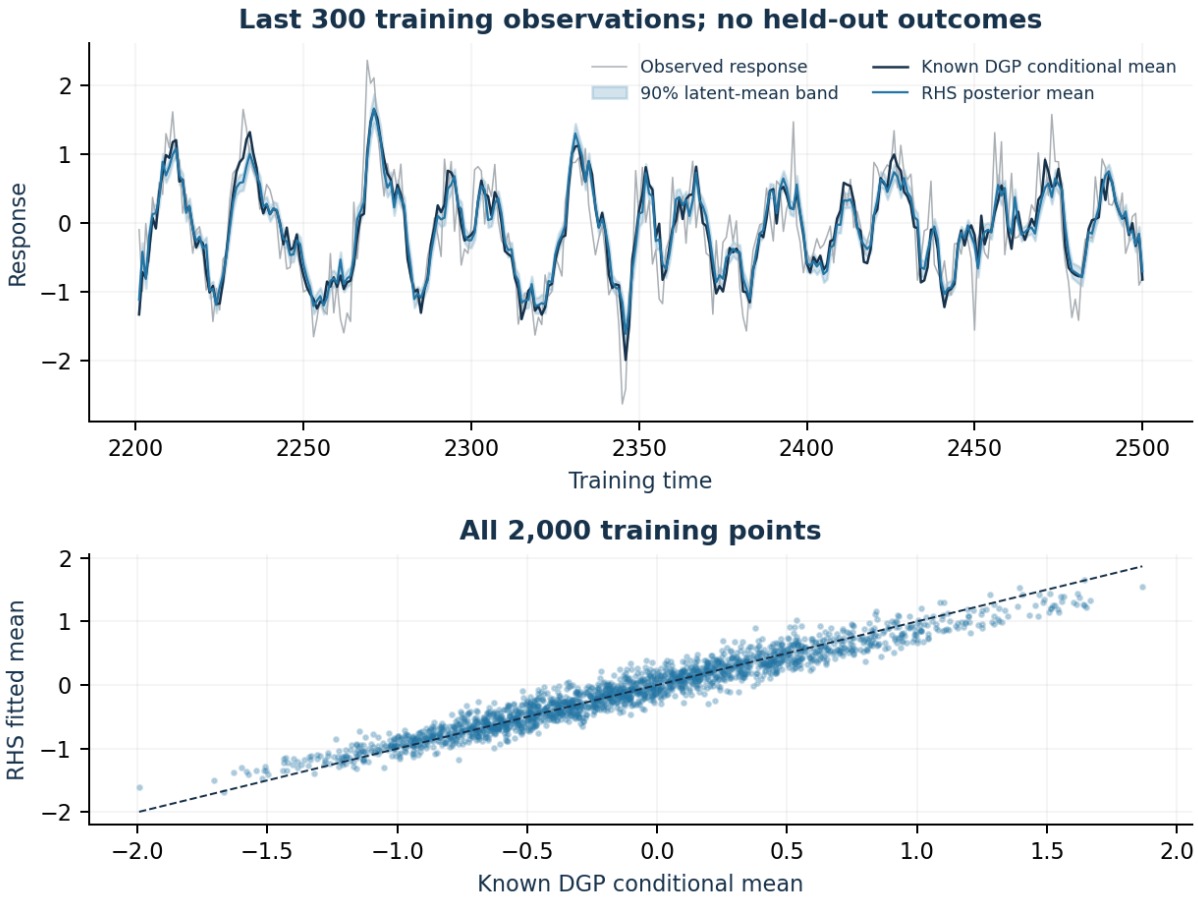
Each architecture is projected separately onto the known one-step conditional mean along observed training history, using an unpenalized SVD fit on rows 501:2500. No held-out observations enter this audit. The feature geometry of all 426 candidates reproduces the frozen bank hash, and nested-space residuals never increase.

Direct lags substantially reduce the per-architecture training approximation error. This is evidence that the added basis can represent useful mean structure. It does not show that the RHS posterior has learned it, that shrinkage is adequate, or that recursive forecasts improve.

The period60 probe is privileged. It tests a calendar space also present in the automatic model support, but supplies no period information to the automatic fit. The plotted weights are architecture priors, not posterior probabilities. A Bayesian averaged mean combines spaces; these per-model bounds are not lower bounds on its error.

All designs are numerically full rank under the registered SVD threshold. Median condition numbers are about 2022-4269; the maximum is about 1.9e8. Correlated feature directions can make individual coefficients difficult to interpret even when fitted functions are stable.

Retained signal, remaining mean bias



Training-only diagnostic	Response units
RHS mean vs known conditional mean	0.1534
RHS mean vs observed response	0.4754
Realized innovations: observed minus true mean	0.4604
Weighted unpenalized fits vs true mean	0.1822
Weighted privileged projections vs true mean	0.1231

The fitted/true mean correlation is 0.9696. Shrinkage improves the descriptive conditional-mean error relative to unpenalized fits of the same stored architecture mixture. The privileged truth projections show additional representational room; neither projection is a fitted forecasting alternative or an error bound for Bayesian averaging.

The latent-mean bands omit observation noise. Their descriptive within-training 90% truth coverage is 54.4%, indicating substantial remaining mean discrepancy. This is not predictive interval coverage or independent repeated-sample coverage. These are 8,192 stored joint states from the first individually passing level; full posterior qualification is pending.

Feature and common-scale audit

Executed standardization check	Result
All 426 reservoir and direct-lag readouts	PASS
All 101 periods, F2 and F2flat	PASS; calendar matrices identical
Largest absolute feature mean	8.76e-14
Largest unit sample-SD error	6.66e-16
Intercept	Constant; proper prior; exempt
Dimension division	None

The audit reads the actual inference Gram matrices and calendar designs. Every slope is centered and scaled using training rows 501:2500. Reservoir input/response conditioning retains the frozen first-500 transformation. These verified fits do not require a standardization redo.

Shared RHS quantities	Four-chain result
All 90 lag coefficients, log locals, log prior variances	270 monitors
Largest rank/folded R-hat	1.001325
Minimum bulk / tail ESS	7,136 / 6,332
Failed monitors	0

These supplementary checks use 2,048 matching joint states per chain. They cover all common lags rather than a few selected coordinates. Reservoir-specific quantities change dimension; their exploration remains subject to the full-trace architecture and cross-budget checks. Finite diagnostic screens cannot establish convergence of every posterior functional.

The independent 512k level was durably verified at 320k total sweeps per chain: 128k warmup and 192k retained. That pause is not a completed 512k fit or an independent shorter-budget fit. Subsequent continuation is recorded in invocation/resource files; it cannot promote the posterior until final validation.

Native GP: completion is not qualification

16-particle seed	Observed RMSE	True-mean RMSE	Particle ESS	Max weight
7366021	0.3621	0.2527	7.28	0.258
7366022	0.4240	0.2562	1.08	0.962
7366023	0.4450	0.3011	2.32	0.634
7366024	0.4008	0.2402	10.86	0.193

All four independent native depth-four, 2,000-row runs completed and their serialized posteriors, training-only inputs, stage records and resource exits were retrieved and verified. Independent streamed Gaussian algebra reproduces all native training probes, with maximum error 7.5e-14. No forecast has been issued from these runs.

The GP fits have smaller observed training residuals than RHS, but larger errors against the DGP conditional mean: 0.240-0.301 versus 0.153 for RHS. This is consistent with some noise fitting. The comparison is descriptive on one training realization; it does not establish forecasting performance, convergence, or superiority.

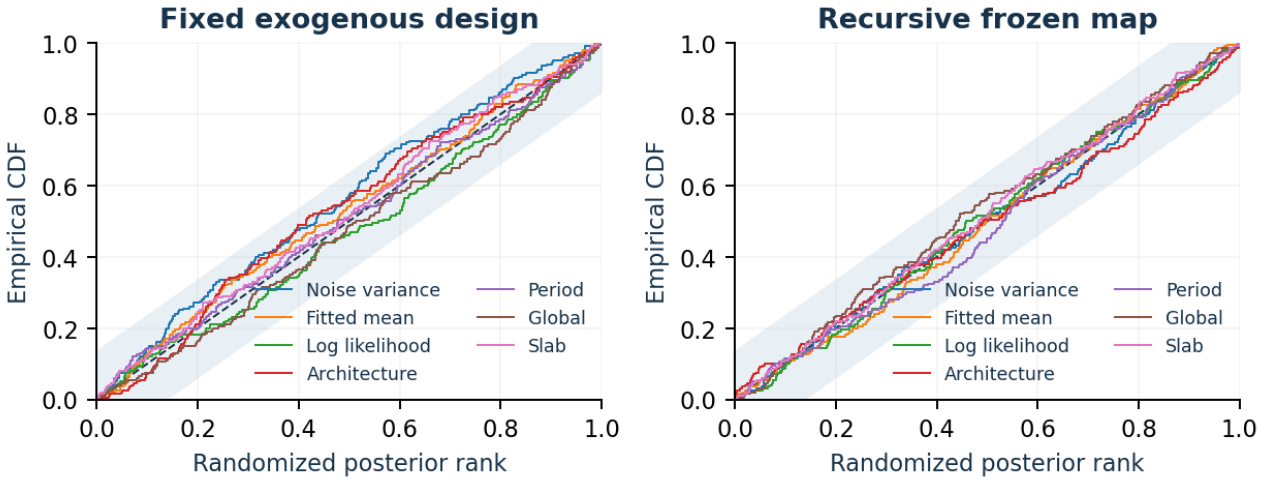
The four GP evidence estimates span 365.3 log units. Correct evidence-weighted aggregation has effective replica count 1.0, with virtually all weight on one run. This fails the registered mode/budget review. Four finished jobs are insufficient evidence of reliable Bayesian averaging.

A concrete accounting defect was found in the pinned native API: it returns the evidence accumulated before the last resampling. Native SMC leaves final particle weights unresampled. Correct pooling adds logsumexp(final log weights) - log(particle count), as in Gen.log_ml_estimate. The postprocessing now reproduces Gen exactly; no completed forecast used the old partial field.

Independent normalized-prior integration checks the complete depth-one continuous leaf target against 64 native SMC replicas per arm (128 particles), with and without native rejuvenation. All 10 unnormalized-measure comparisons pass. This validates small-target normalizer accounting, not depth-four exploration.

Calibration scope and actual execution status

Calibration	Executed result / limitation
Fixed-design ridge	200 cases; rank envelope passed; minimum monitored spectral ESS 810.1; 0 cases below 400.
Fixed-design RHS, initial budget	200 cases; rank envelope passed; minimum monitored spectral ESS 66.5; 136 cases below 400.
Fixed-design RHS, 24k budget	200 cases; rank envelope passed; minimum monitored spectral ESS 1080.8; 0 cases below 400.
Recursive feasibility pilot	12 cases; simultaneous rank bound 0.549, observed deviation 0.540; 9 cases below ESS 400. Too small and computationally unresolved.
Recursive constant, 24k	200 cases; rank envelope passed; minimum ESS 301.5; 1 cases below 400.
Recursive sv, 24k	200 cases; rank envelope passed; minimum ESS 620.2; 0 cases below 400.
Recursive constant, recovery	Same 200 cases; case 11 recomputed at 96k retained after ESS-only selection. Rank envelope passed; minimum monitored ESS 643.5; no cases below 400.



Fixed-design SBC holds candidate matrices exogenous. Recursive-model SBC freezes weights, projections, input/response transforms and readout scalers on an independent prefix, then generates and reconstructs only causal paths. Neither validates unconditional calibration of a production pipeline that estimates its transformations from each observed response series.

The pilot is retained, not pooled into a larger batch or counted as confirmation. Calibration and confirmation seed namespaces are recorded separately. No NHS1 confirmatory series has been generated or unblinded. A seed-block assignment change before confirmation is documented in seed_namespace_decision.json.

Fixed-design and recursive RHS calibration use tau0=0.2 and slab_scale=1, slab_df=4. Their priors and independently frozen feature maps differ from the production fit and their conclusions have that scope.

The initial and larger fixed-design budgets use the same 200 prior-predictive cases. Recursive recovery also adds no independent cases. The fixed-design noise-variance ranks have the largest deviation (0.117 versus the registered simultaneous bound 0.135); passing an envelope is not proof of exact calibration.

Runtime, memory and engineering checks

Observed process	Wall (s)	Aggregate PSS (GiB)	Exit / status
Original full-bank fork	1,176.3	unreliable early scope	1
Exact cache, F1 32k	662.3	3.83	0
Exact cache, F1 128k	2,803.3	4.63	0
Exact local move, F1 32k	804.7	3.84	0
Joint scale/local policy, F1 32k	834.8	3.84	0
Joint F1 256k continuation	4,057.1	5.02	0
Recover compact forecast maps	45.2	3.40	0

Process wall time includes startup, compilation/cache loading and saving. Fit checkpoints separately record inference elapsed time. Earlier aggregate PSS sampling missed descendants; those values are not used as measured total memory. Corrected records sample process-tree PSS and explicitly label environment-wide anonymous memory, which includes concurrent jobs.

All 426 numeric reservoir maps preserve before/after transitions exactly within each serial and batched backend. Last training features differ by at most 1.8e-13. Compact maps occupy 7.82 MB. They remove unused construction closures, not weights or candidate models.

An H_last/H_all integration defect was corrected before any new benchmark forecast. It affected the new adapter, not the already frozen historical forecast results. The earlier OOM, corrupted derived cache rejection, incomplete GP run and failed launcher remain part of the cost/evidence ledger.

Paired small-target trace-storage benchmark: median 11.401 s with data frames, 8.897 s with matrices, preserving full traces and matching joint states bit for bit. This is not a full-study speedup claim.

Forecast performance remains separately scoped: the synthetic reservoir-64/H30/192-outer/8-inner batch benchmark was 2.359 s versus 0.285 s, maximum path difference below 1e-13. No new full rolling-panel forecasting runtime is available.

Exact computation and interruption recovery

Paired trial	Uncached (s)	Cached (s)
1	52.030	29.476
2	49.976	30.306
3	46.513	31.434

Three same-seed full-bank trials preserve all recorded traces, matching coefficient/local/volatility states, fitted probes and architecture masses bit for bit. Median sampling time is 49.976 s without the fixed statistics cache and 30.306 s with it; preparing all 426 F1 entries took 1.859 s. Each trial uses 500 warmup and 2,000 retained sweeps. These paired computations are not independent DGP replications or a complete-study speedup.

The optional cache stores fixed unweighted X-transpose-X and X-transpose-y values under a numeric-payload memory bound. Partial caching falls back to exact computation and retains every model/period. It does not change the prior, target, posterior weights or random updates. Weighted SV statistics are recomputed as covariance changes.

The first fresh 256k invocation disappeared before completed chains or an exit/resource record were observed. Its cause and final iteration are unknown. It contributes neither a completed posterior nor an invented runtime total. This interruption exposed the need for checkpoints within each long chain.

Recovery now saves all Markov variables, auxiliary scales, retained draws, probes, counters and RNG state at complete sweep boundaries. A separate four-process C++/grouped-RHS/seasonal-SV experiment pauses at sweep 47 and finishes the same 120-sweep budget; every recorded state and counter matches uninterrupted computation exactly.

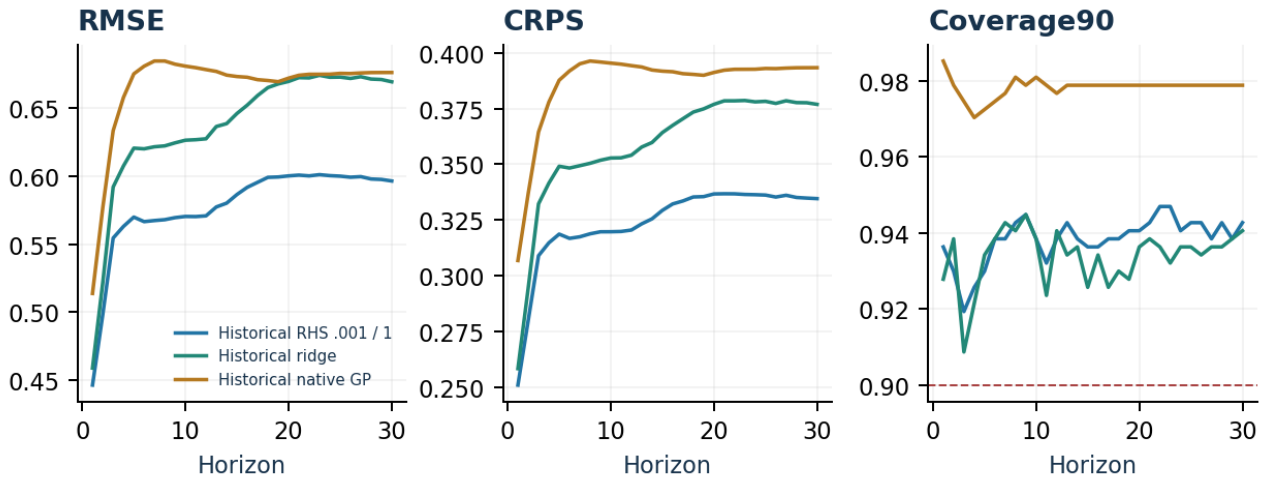
Resume verifies the target, kernel, full budget, source, runtime, thread settings and payload checksum. Box-Muller RNG is rejected because its hidden cached normal is absent from .Random.seed. Bounded pauses are explicitly incomplete fits and do not authorize forecasts.

The earlier bounded recovery audit recorded: All four full-bank F1 chains durably paused at sweep 8,000, restarted in fresh processes and paused again at sweep 16,000. Both invocations exited zero; payload and file checksums and cumulative proposal counters verify. Combined observed process wall time is 249.97 s, maximum sampled aggregate PSS 4.36 GiB. These are warmup-only states: zero retained posterior draws, no production qualification and no forecasts. The earlier unmeasured interrupted invocation remains separate.

The subsequent full continuation completes all four chains with exit 0. Summing the three observed process invocations gives 71.78 active minutes, including repeated startup/checkpoint work and excluding pauses between invocations. Peak process-tree PSS is 5.02 GiB; sampled environment anonymous memory includes simultaneous audits. The earlier unknown-exit invocation remains uncredited.

No new full rolling-origin forecast or independent confirmation result is produced by these engineering checks. The model qualification thresholds remain unchanged.

Historical forecast context - unchanged



Historical model	RMSE	MAE	CRPS	NLL	50% cover / width	90% cover / width
Historical RHS .001 / 1	0.579	0.447	0.324	0.887	0.593 / 0.907	0.938 / 2.211
Historical ridge	0.639	0.498	0.359	0.978	0.575 / 0.968	0.934 / 2.324
Historical native GP	0.666	0.524	0.387	1.103	0.666 / 1.260	0.978 / 3.073

These scores are the original completed NHS1 reference, not outcomes of the new augmented or volatility models. Each row pools 471 rolling origins and 30 horizons. The 14,130 pairs are dependent observations from one realization; they are not 14,130 independent simulation replications.

High pooled coverage can conceal excess width and regime-dependent undercoverage. Visual failure to track every realized spike also includes irreducible innovation uncertainty. The new work must separately establish computational adequacy and then compare conditional means, proper scores and calibration. None of the historical rankings proves that a new model is superior.

Unfinished gates and reproducible handoff

Gate	What is still required
Posterior qualification	Two successive qualified fresh budgets, targeted mode checks, probability precision and cross-budget stability under frozen computational controls.
Mean-family selection	Rebuild prefix banks and refit scalers at cutoffs 1500 and 2000; qualified frozen-fit rolling validation; select by registered CRPS rule.
Native AutoGP	Complete full-size independent same-prior runs and particle/move-budget sensitivity. Small feasibility runs and an interrupted full run do not qualify it.
Conditional variance	Recursive constant/SV calibration has resolved within its frozen-map scope. Production-chain qualification and conditional particle-filter budget checks remain.
Prediction	Outer/inner Monte Carlo panel; full 471-origin, 30-horizon comparisons with RMSE, MAE, CRPS, NLL, intervals and calibrated runtimes.
Independent confirmation	Frozen 30-realization mean/GP block, separate variance block and registered stress cells. None can be replaced by overlapping origin-horizon pairs.

Checkpoints preserve completed numeric states, traces, case records, immutable executed sources and failure logs. restore_checkpoints.py verifies archive ancestry, every part and every extracted member. Large construction/inference banks are referenced by checksums and deterministic rebuild scripts rather than duplicated in each fitted object.

Primary evidence: MATH.md; execution.json; development_registry.json; independent-reference check CSV/JSON files; F1 checkpoint-validation and qualification records; resource records; versioned checkpoint manifests. The report input manifest supplies their exact SHA256 values.

The appropriate conclusion at this checkpoint is an implemented and increasingly validated statistical extension with unresolved production and predictive gates. It is not a completed forecast comparison or a superiority claim.